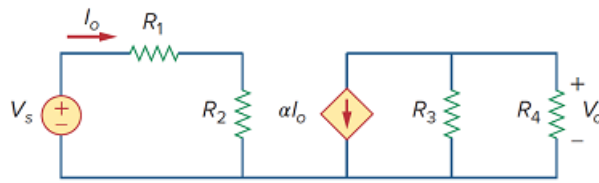


Problem

For the circuit in Fig. 2.88, find V_o/V_s in terms of α , R_1 , R_2 , R_3 , and R_4 . If $R_1 = R_2 = R_3 = R_4$, what value of α will produce $|V_o/V_s| = 10$?

Figure 2.88:



Step-by-step solution

Step 1 of 4

Consider the following circuit diagram:

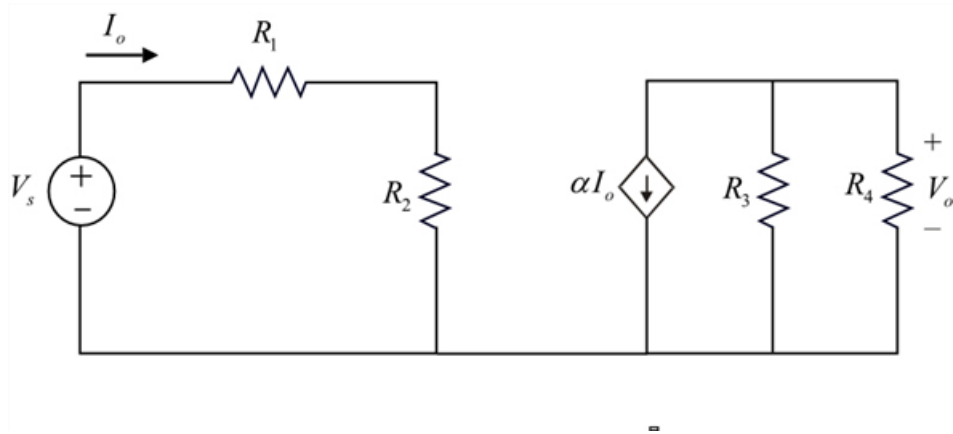


Figure 1

Step 2 of 4

Find the current I_o flows through the loop 1 by applying Kirchhoff voltage law, as shown below

Figure 2

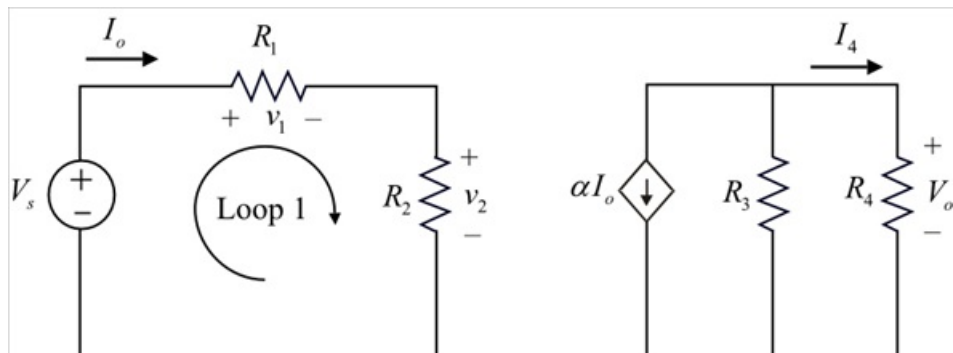


Figure 2

Step 3 of 4

Apply Kirchhoff's voltage law around the loop1.

$$V_s - v_1 - v_2 = 0$$

Express above equation in terms of current I_o .

$$V_s - R_1(I_o) - R_2(I_o) = 0$$

Rearrange the terms

$$I_o(R_1 + R_2) = V_s$$

$$I_o = \frac{V_s}{R_1 + R_2} \dots\dots (1)$$

Apply current division rule to find the current through resistor R_4 .

$$I_4 = -\alpha I_o \left(\frac{R_3}{R_3 + R_4} \right)$$

Apply Ohm's law to resistor R_4 .

$$V_o = I_4(R_4)$$

Substitute $-\alpha I_o \left(\frac{R_3}{R_3 + R_4} \right)$ for I_4 in the above equation.

$$\begin{aligned} V_o &= -\alpha I_o \left(\frac{R_3}{R_3 + R_4} \right) (R_4) \\ &= -\alpha I_o \left(\frac{R_3 R_4}{R_3 + R_4} \right) \end{aligned}$$

From equation (1), substitute $\frac{V_s}{R_1 + R_2}$ for I_o in the above expression.

$$\begin{aligned} V_o &= -\alpha \left(\frac{V_s}{R_1 + R_2} \right) \left(\frac{R_3 R_4}{R_3 + R_4} \right) \\ \frac{V_o}{V_s} &= \frac{-\alpha R_3 R_4}{(R_1 + R_2)(R_3 + R_4)} \end{aligned}$$

Therefore $\frac{V_o}{V_s}$ in terms of α, R_1, R_2, R_3 and R_4 is $\boxed{\frac{-\alpha R_3 R_4}{(R_1 + R_2)(R_3 + R_4)}}$.

Step 4 of 4

Assume $R_1 = R_2 = R_3 = R_4 = R$.

Consider $\frac{V_o}{V_s}$ expression.

$$\frac{V_o}{V_s} = \frac{-\alpha R_3 R_4}{(R_1 + R_2)(R_3 + R_4)}$$

Equate $\left| \frac{V_o}{V_s} \right|$ to 10 to get the α value at condition $R_1 = R_2 = R_3 = R_4 = R$.

$$\begin{aligned} \left| \frac{-\alpha R_3 R_4}{(R_1 + R_2)(R_3 + R_4)} \right| &= 10 \\ \frac{\alpha R^2}{(2R)(2R)} &= 10 \quad \text{since } R_1 = R_2 = R_3 = R_4 = R \end{aligned}$$

$$\frac{\alpha}{4} = 10$$

$$\alpha = 40$$

Therefore the value of α is $\boxed{40}$ to produce $\left| \frac{V_o}{V_s} \right| = 10$.